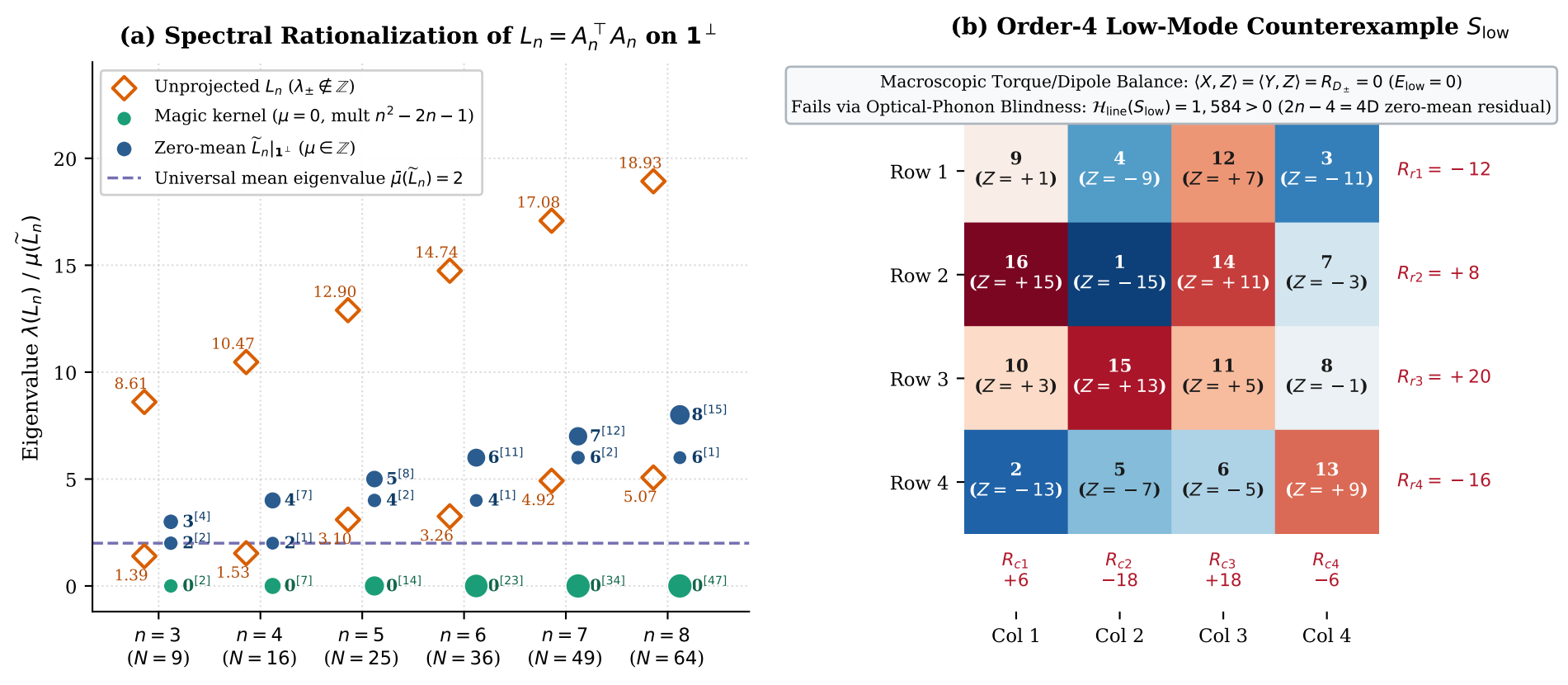

(a) Spectral Rationalization of $L_n = A_n^\top A_n$ on $\mathbf{1}^\perp$
Unprojected L_n ($\lambda_\pm \notin \mathbb{Z}$)
Magic kernel ($\mu = 0$, mult $n^2 - 2n - 1$)
Zero-mean $\widetilde{L}_n|_{\mathbf{1}^\perp}$ ($\mu \in \mathbb{Z}$)
Universal mean eigenvalue $\bar{\mu}(\widetilde{L}_n) = 2$
Eigenvalue $\lambda(L_n)$ / $\mu(\widetilde{L}_n)$
0
5
10
15
20
8.61
10.47
12.90
14.74
17.08
18.93
1.39
1.53
3.10
3.26
4.92
5.07
$n = 3$ ($N = 9$)
$n = 4$ ($N = 16$)
$n = 5$ ($N = 25$)
$n = 6$ ($N = 36$)
$n = 7$ ($N = 49$)
$n = 8$ ($N = 64$)
(b) Order-4 Low-Mode Counterexample S_{low}
Macroscopic Torque/Dipole Balance: $\langle X, Z\rangle = \langle Y, Z\rangle = R_{D_\pm} = 0$ ($E_{\text{low}} = 0$)
Fails via Optical-Phonon Blindness: $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1,584 > 0$ ($2n - 4 = 4$D zero-mean residual)
Row 1
Row 2
Row 3
Row 4
9 ($Z = +1$)
4 ($Z = -9$)
12 ($Z = +7$)
3 ($Z = -11$)
16 ($Z = +15$)
1 ($Z = -15$)
14 ($Z = +11$)
7 ($Z = -3$)
10 ($Z = +3$)
15 ($Z = +13$)
11 ($Z = +5$)
8 ($Z = -1$)
2 ($Z = -13$)
5 ($Z = -7$)
6 ($Z = -5$)
13 ($Z = +9$)
$R_{r1} = -12$
$R_{r2} = +8$
$R_{r3} = +20$
$R_{r4} = -16$
R_{c1} +6
R_{c2} −18
R_{c3} +18
R_{c4} −6
Col 1
Col 2
Col 3
Col 4